

Acute Lymphoblastic Leukaemia Diagnosis using EfficientNet B3 Transfer Learning Model

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Abstract— Acute Lymphoblastic Leukaemia (ALL) is a kind of cancer that affects the leukocytes in the bone marrow and blood. Improved treatment results and patient survival depend on the early identification of ALL. A computer-aided diagnosis method for ALL detection was created in this work using a dataset of 15,135 pictures from 118 individuals with two labelled classifications (normal cell and leukaemia blast). For classification, the EfficientNetB3 model of transfer learning was used. The training (70%), validation (12%), and testing (18%) sets were created by randomly dividing the dataset. To enhance the dataset size and avoid overfitting, data augmentation techniques were used on the training set. With a sensitivity of 97.58% and a specificity of 92%, the model's accuracy on the testing set was 95.75%. These findings show the potential of deep learning systems for ALL early diagnosis. Overall, this work shows that leukaemia blast cells in peripheral blood smears can be effectively classified using the EfficientNetB3 transfer learning model. The creation of automated image-based diagnostic tools can assist in shortening the manual inspection process and save time and money while also producing more accurate and repeatable findings. This would help in reducing preventable deaths and neonatal mortality while improving the healthcare sector.

Keywords—Acute Lymphoblastic Leukaemia, CNN, EfficientNetB3, Transfer Learning Model

I. INTRODUCTION

Acute Lymphoblastic Leukaemia (ALL) is a kind of cancer that affects the leukocytes [1] in the bone marrow and blood. ALL, which makes up around 25% of paediatric malignancies, is the most prevalent kind of childhood cancer. While ALL is a commonly occurs in childhood, it can be diagnosed in adults as well. The efficacy of ALL treatment depends on the disease's early identification. By identifying leukaemia blast cells in blood cultures under a microscope, a pathologist can detect ALL using the traditional approach. Unfortunately, this approach is laborious, arbitrary, and time-consuming. Recently, automated leukaemia blast identification using microscopic pictures of blood smears has demonstrated encouraging results using deep learning algorithms. The goal of this study is to detect ALL from 15,135 photos of 118 patients using the cutting-edge deep learning model EfficientNetB3 and classify them into 2 classes: normal cell and leukaemia blast. Making the right treatment decisions for patients depends on accurate diagnosis of ALL. Deep learning [2] models like the EfficientNet have demonstrated significant promise in recent years in terms of increasing the precision of cancer detection and classification. Large datasets may be used by these algorithms to understand intricate patterns and correlations

that can be used to distinguish minute variations between various cancer cell types. In this project, transfer learning with the EfficientNetB3 model was utilized to detect and diagnose leukaemia cancer based on microscopic pictures of blood cells. The ImageNet dataset, which includes millions of pictures from many classifications, was used to pre-train the CNN EfficientNetB3. A pre-trained model is utilized as an initial point for a new assignment using the deep learning approach known as transfer learning as opposed to building a model from beginning. This enables us to use the know-how and experience that the pre-trained model, which has been trained on big datasets, has already acquired to enhance the accuracy of our model with a smaller dataset. A class of CNNs called EfficientNet was created primarily for the effective use of computing resources. These models have attained cutting-edge performance on a number of tasks associated with computer vision, including segmentation [3], object identification, and picture classification. Micrographs of blood cells taken from individuals with ALL make up our dataset. Each image was tagged with the appropriate label, and this knowledge was used to train the model to correctly categorize the photos. The EfficientNetB3 model was first adjusted using a leukaemia dataset from Kaggle. The pre-trained model's weights were changed during fine-tuning in order to better suit the unique purpose of this research. The first few layers of the model, that have already learnt generic traits, were frozen while the final few layers were retrained to identify ALL. With the last few layers learning leukaemia-specific properties, advantage can be taken of the pre-trained model's understanding of generic features like edges, textures, and forms. Following model optimization, the model's performance was assessed using common classification metrics including recall, accuracy, and precision, and F1-Score. The precision and recall metrics indicate, respectively, the percentage of true positive forecasts and the percentage of real positives that are properly detected by the model, while the accuracy parameter reflects the ratio of correctly classified images. The ultimate objective of this research is to create an acute lymphoblastic leukaemia classification model that is very accurate and effective and that medical personnel may utilize to enhance patient outcomes. The proper treatment strategy for each patient may be determined with the aid of an accurate diagnosis of leukaemia, possibly improving outcomes and quality of life. We can create more efficient treatment regimens and eventually improve patient outcomes by better understanding the distinctions between normal cells and ALL affected cells to detect and diagnose ALL accurately.

The major contributions of this paper are:

- The goal of this study is to reliably diagnose acute lymphoblastic leukaemia cancer by combining the EfficientNetB3 model with deep learning and transfer learning techniques.
- An ALL dataset from Kaggle was employed consisting of 15135 images from 118 patients. The dataset consisted of images of 2 labelled classes: Normal Cell and Leukaemia Blast.
- The implemented model shows a high accuracy of 95.75% for the purpose of detecting ALL from blood smear images. The Sensitivity of the model is 97.58% and the Specificity of the model is 92%.

The entire paper has been divided into these six sections: We discuss several relevant studies on leukocyte categorization in section II. The proposed methodology is outlined in Section III. Part IV describes the performance of the suggested approach. The manuscript is concluded in Section V.

II. RELATED WORK

Anwar et al. [4] described an automated diagnostic method that used a CNN model to identify ALL. The model used labelled microscopic blood smear images to detect the malignant leukaemia cells. The study used data from the Acute Lymphoblastic Leukaemia Imaging DataBase (ALL-IDB) and applied a variety of data augmentation approaches to improve the amount of training data. The model was evaluated on the remaining 221 photos after being trained on 515 images using a fivefold validation procedure, reaching an accuracy of 95.54%. Deep CNNs were used by Mondal et al. [5] to automate the ALL detection process from tiny cell pictures. To suggest a superior ALL cell classifier, the weighted ensemble of several deep CNNs were examined. Using the relevant metrics, such as accuracy, F1-score, AUC, and kappa values, the weights for the ensemble candidate models were calculated. Different data augmentations and pre-processing were utilized for attaining a greater generalization of the network. The experiments were conducted comprehensively by utilizing the C-NMC-2019 ALL dataset. The suggested weighted ensemble model produced a weighted F1-score of 88.6% and an accuracy of 86.2%. To help in the diagnosis of ALL, Jiang et al. [6] presented the ViT-CNN ensemble model to identify cancer cells pictures and normal cells images. To get better classification results, the ViT-CNN ensemble model was utilized to extract the characteristics of cell pictures in two totally distinct approaches. A difference enhancement-random sampling (DERS) data enhancement method was also suggested to create a new balanced data set from the unbalanced data set used in the article, and the symmetric cross-entropy loss function was employed to lessen the effect of noise in the data set. For automatic identification of acute lymphoblastic leukaemia and categorization of its subtypes into 4 classes—L1, L2, L3, and Normal—Shafique et al. [7] used the pre-trained CNN, AlexNet. As opposed to training a CNN from scratch, the pre-trained AlexNet TL model was used that had been adjusted to the input data set. Final levels of the pre - trained model were changed with new layers which could categorize the input images into four categories. To minimize overtraining, augmentation of the input data was carried out. In order to evaluate the

performance across various colour colour photographs, comparison of data sets with various colour models was also implemented. Ullah et al. [8] proposed a CNN-based method for detecting ALL by making use of medical pictures. The presented approach involved extracting higher quality feature information from the image dataset using VGG16 and an attention module termed Efficient Channel Attention (ECA), which improved feature representation and classification outcomes. The suggested technique demonstrated how the ECA module aids in reducing morphological resemblance between pictures of healthy and cancerous cells. The amount and quality of training data were additionally increased by using a variety of augmentation strategies. Based on subject-level variability, the C-NMC dataset was split into seven folds. According to experimental findings, the presented CNN model was able to extract deep features with an accuracy of 91.1%. In order to address the problem of CNNs requiring enormous databases for adequate training and produce more accurate ALL detection, Das et al. [9] designed an effective deep CNNs architecture. A unique probability-based weight factor was presented in this method's proposal, and it was crucial in effectively combining MobilenetV2 with ResNet18 while keeping the advantages of both systems. Using the available benchmark datasets ALLIDB1 and ALLIDB2, its performance was verified. A novel approach based on histopathological transfer learning was put out by Genovese et al. [10] for the identification of ALL. The approach trained a CNN using a histopathology database to identify tissue types before fine-tuning it on the ALL database to determine the presence of lymphoblasts. In contrast to CNNs pretrained on ImageNet, a multilabel dataset with a significantly higher number of samples and classes was taken into account as the histopathology database. This allowed CNNs to learn basic characteristics for histopathology image analysis and, consequently, allowed them to perform more effective transfer learning. Several CNNs were taken into account when the approach was assessed using a publicly accessible ALL database. A novel Bayesian-based optimized CNN was presented for the identification of ALL in microscopic smear pictures by Atteia et al. [11]. The design of the proposed CNN and its hyperparameters were tailored to input data using the Bayesian optimisation technique in order to improve classification performance. In order to find the set of network hyperparameters that minimises an objective error function, the Bayesian optimisation approach employs an educated iterative search process. Using a hybrid dataset created by combining two open-source datasets, the suggested CNN was trained and verified. To improve classification performance, data augmentation was used to further feed the hybrid picture collection. For the test set, the Bayesian search-derived ideal CNN model performed better in image-based ALL classification. Using an automated system that incorporates an object identification algorithm that detects leukemic cells from microscopic blood sample pictures, Khandekar et al. [12] suggested automating the blast cell detection. Cell recognition and cell categorization were both accomplished using the You Only Look Once (YOLOv4) method. As a result, each cell was given a label of either blast cells (ALL) or healthy cells, and the classification was set up as a binary issue (HEM). Images from the ALL IDB1 and C NMC 2019 datasets used as training and testing examples for the Object

Detection algorithm. Rezayi et al. [13] classified leukemic cells from healthy cells in microscopic pictures using a dataset gathered from a CodaLab competition. ResNet-50 and the VGG-16, two well-known deep learning networks, were used. The stored weights weren't used, and the learning parameters and weights were both modified because these two networks had previously been trained using certain provided parameters. Also, six widely used machine learning approaches were created to divide acute lymphoblastic leukaemia into two groups, along with a neural network with ten convolutional layers and two by two max-pooling layers—with strides 2. The ResNet-50, VGG-16, and proposed CNN all had validation accuracies of 81.63%, 84.62%, and 82.10%, respectively.

III. METHODOLOGY

In this paper, an ALL dataset from Kaggle was employed consisting of 15135 images from 118 patients. The dataset consisted of images of 2 labelled classes: Normal Cell and Leukaemia Blast. The pre-trained EfficientNetB3 TL model was implemented for the purpose of classifying the images into 2 classes based on whether leukaemia is detected. Fine-tuning layers were added to the EfficientNetB3 model to obtain better results with higher accuracy. The EfficientNetB3 model along with the fine-tuning layers is highlighted in Figure 1.

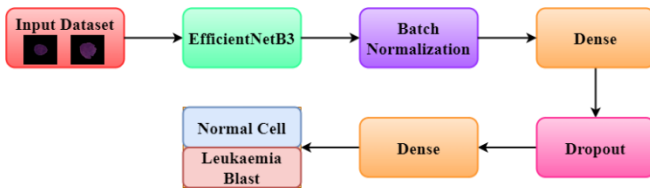
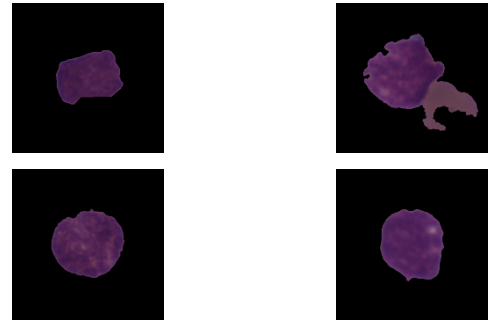
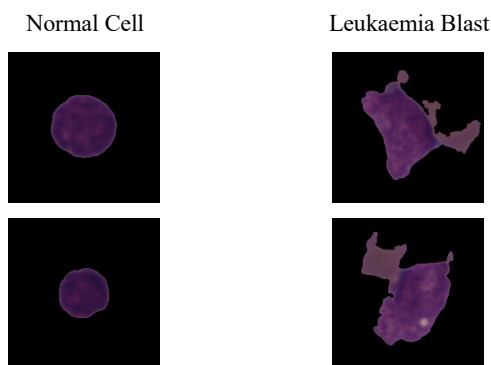


Fig 1. EfficientNetB3 Model with Fine-Tuning Layers for Leukemia Diagnosis

A. Dataset Description

For this study, the dataset was obtained from Kaggle platform and it consists of 15135 blood smear images from 118 patients. The dataset consisted of 2 separate classes: normal cell and leukaemia blast. The dataset was split into three separate divisions of Training, Validation, and Testing sets containing varying number of images. The training (70%), validation (12%), and testing (18%) sets were created by randomly dividing the dataset. Some images of the 2 classes of the dataset are displayed in Table 1.

TABLE I. IMAGES OF INPUT DATASET CLASSES



B. EfficientNetB3

In 2019, the Google research team created the neural network architecture known as EfficientNetB3 [14]. It is a member of the EfficientNet series of models, which aim for greater accuracy while using minimal computational resources. The construction of EfficientNetB3 is intricate and includes a number of methods to enhance the network's efficiency and effectiveness, such as:

1. Compound scaling: To increase accuracy, this method evenly scales up the network's depth, breadth, and resolution parameters.
2. Depth wise separable convolutions: This method divides the typical convolutional process into two pieces, a depth wise convolution and a pointwise convolution, which greatly lowers complexity while retaining accuracy.
3. Swish activation function: It has been demonstrated that this smooth, non-monotonic activation function outperforms more conventional activation functions like ReLU.
4. DropConnect regularization: This method reduces overfitting and enhances generalization by randomly removing connections between neurons during training.

In conclusion, EfficientNetB3 is a highly accurate and efficient neural network capable of being employed in a number of computer vision applications, such as segmentation, object identification, and picture classification.

C. Fine-Tuning Layers

In a neural network, fine-tuning [15] refers to the act of altering or adjusting pre-trained layers to suit a particular job. CNNs are often fine-tuned by retraining the final few layers of the network on a new task, such as object identification or image classification, using a pre-trained network design that has been trained on a large dataset (e.g., ImageNet). During the fine-tuning process, the pre-trained layers are frozen, which means that their weights are set and not modified throughout training. The last few layers, however, are often updated or tweaked to fit to the new job, and their weights are learned using the new data. Since it drastically minimizes the quantity of data and training time needed to attain exceptional performance on a new task, fine-tuning is the preferred deep learning strategy. By exploiting pre-trained networks that have previously learnt a rich collection of characteristics from a huge dataset, fine-tuning allows us to transfer knowledge from one job to another and develop effective models with less training data.

In this study, Batch Normalization, Dropout, and Dense layers were utilized for fine-tuning the EfficientNetB3 TL model.

D. Model Summary

Table 2 highlights the summary of the model implemented in this study consisting of EfficientNetB3 TL model and the fine-tuning layers. The input image size and output image size highlight the size of the image before and after processing from the corresponding layer respectively.

TABLE II. IMAGES OF INPUT DATASET CLASSES

Layer Name	Input Image Size	Output Image Size	Activation Function	Parameters
Input Layer	NA	224 x 224 x 3	-	0
EfficientNetB3	224 x 224 x 3	1536	-	10783535
Batch Normalization	1536	1536	-	6144
Dense	1536	256	ReLU	393472
Dropout	256	256	-	0
Dense	256	2	Softmax	514

In one of the Dense layers, the ReLU (Rectified Linear Unit) activation function was employed. ReLU is used because it allows for non-linearity, which is necessary for the neural network to understand complex patterns and connections in the input. It leads to sparse activations, which show that only some of the neurons are stimulated for a given input. By doing this, overfitting may be avoided and less information processing is required. ReLU can mitigate the vanishing gradient problem that deep neural networks encounter. As a result of the gradients getting too small during backpropagation, the network is unable to learn. ReLU permits the neurons to output non-zero values, preventing the gradients from being too small. Softmax was employed for the final Dense layer. Since it is an efficient technique that can be used for multi - classification tasks when there are more than two output classes, it is ideal to use in the final output layer. It can operate with any number of classes and build a probability distribution over each one, so the output value's total is always 1. When utilizing a neural network for classification tasks, where the purpose is to predict the likelihood that an input belongs to each class, this is helpful.

The model consists of 11,093,290 parameters that are trainable and 90,375 parameters that cannot be trained. Thus, the total number of parameters are 11,183,665.

IV. RESULTS

The results of the EfficientNetB3 model implementation with fine-tuning layers are highlighted in this section in the form of Loss Analysis Plot, Accuracy Analysis Plot, and

Confusion Matrix. The model's performance was assessed based on its capability of accurately classifying the input dataset images into 2 classes. Common classification metrics that were derived from the Confusion Matrix were used to assess the model's performance.

A. Loss Analysis

The loss sustained during the implementation of the model for both the training and validation sets is shown in Figure 2.

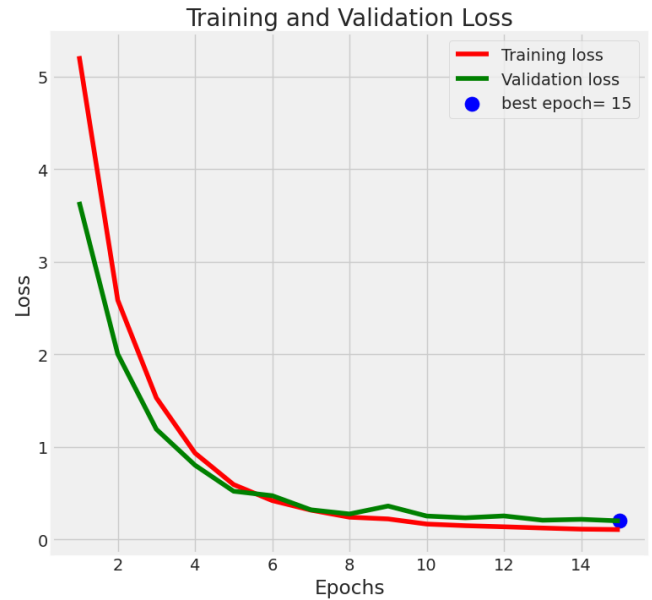


Fig 2. Loss Analysis Plot

As seen in Figure 2, the loss sustained during implementation of the model is very high during initial epochs of model implementation. The loss then exponentially decreases in both the cases and reaches the lowest at epoch 15. The final value of training loss is 0.107 and the final value of validation loss sets at 0.2.

B. Accuracy Analysis Plot

The accuracy attained during implementation for both the training and validation sets is shown in Figure 3.

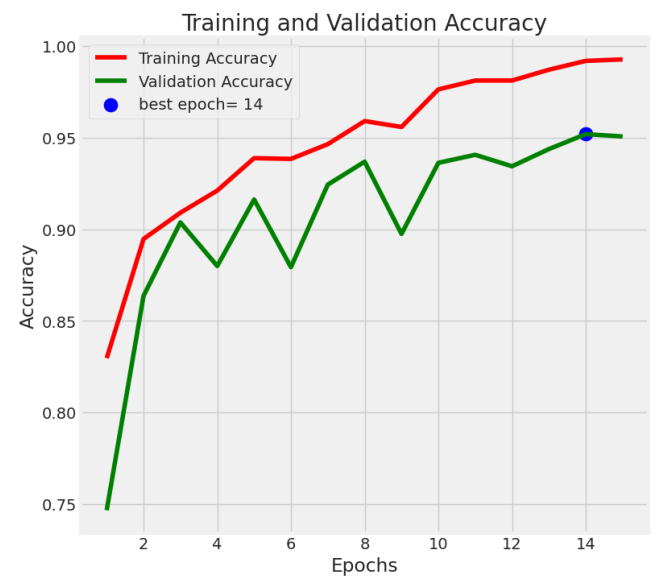


Fig 3. Accuracy Analysis Plot

As seen in Figure 3, the training accuracy attained during implementation of the model starts below 85% and increases gradually to attain a good training accuracy of 99.26% in the final epoch. The validation accuracy during the first epoch is very low, being below 75% and then exponentially increases for the 2nd and 3rd epochs. The accuracy then constantly shows a high-low pattern increasing and decreasing before increasing in a stable manner in the final epochs of model implementation to attain a final validation accuracy of 95.06%.

C. Confusion Matrix

After receiving data as input, a model generates results in the form of probabilities. As a consequence, a confusion matrix [16] is used to evaluate the efficacy of a CNN model. In this table-like presentation, there are countless combinations of expected and actual numbers. As a result, figuring out performance indicators like accuracy is significantly easier. As a result, a confusion matrix is used to summarise a model's performance for a certain dataset. Figure 4 depicts the confusion matrix that is produced by using the obtained probabilities.

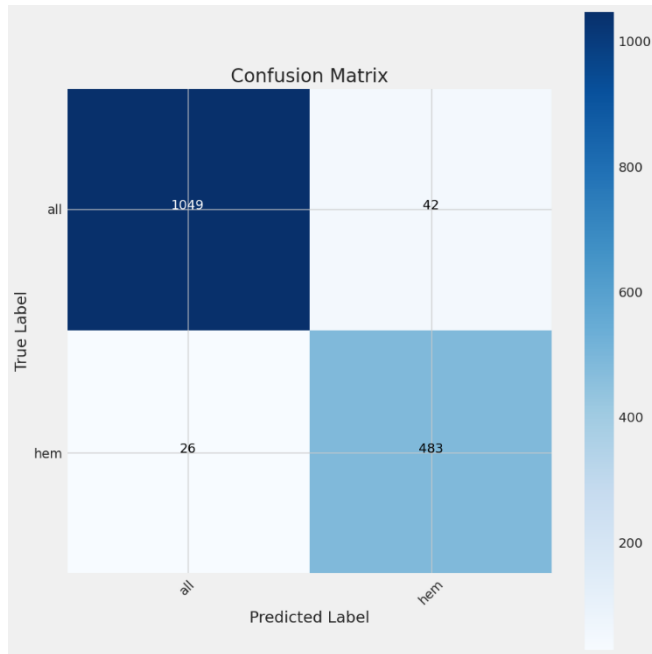


Fig 4. Confusion Matrix

In Figure 4, the 'all' label is used to signify the images for detection of ALL and the 'hem' label is used to signify the normal cell images. The Confusion Matrix is employed to assess the efficacy of the EfficientNetB3 model implemented with fine-tuning layers for the detection of ALL from blood smear images. The model's efficacy is evaluated using common classification metrics including Accuracy, Precision, F1 Score, and Recall.

D. Confusion Matrix Parameters

Table 3 displays the performance metrics [17] derived from the Confusion Matrix that can be utilized to assess the performance of the implemented model for detection of ALL. The calculated performance metrics are Precision, Recall, F1-Score and Accuracy.

TABLE III. CONFUSION MATRIX PARAMETERS

	Precision	Recall	F1 Score	Accuracy
ALL	0.98	0.96	0.97	95.75%
Normal	0.92	0.95	0.93	

Thus, as seen from Table 3, the implemented model shows a high accuracy of 95.75% for the purpose of detecting ALL from blood smear images. The Sensitivity of the model is 97.58% and the Specificity of the model is 92%.

CONCLUSION

In conclusion, transfer learning has demonstrated promising results in the creation of an effective and precise model for the identification of Acute Lymphoblastic Leukaemia. For the classification of blood sample pictures, the EfficientNetB3 architecture, which has been customised with fine-tuning layers, has demonstrated that it is a potent tool. With a remarkable accuracy of 95.75 %, the model was trained using a dataset of 15135 blood sample images from 118 individuals. This high percentage of accuracy shows that the model is a trustworthy and effective tool that can help doctors diagnose ALL. Early and accurate identification are essential for improving patient outcomes since ALL is becoming more common each year. The creation of automated technologies like these can facilitate diagnosis in a shorter amount of time, enabling early treatment and raising overall survival rates. The accomplishment of this project demonstrates the potential of deep learning models in the field of medical diagnostics and stimulates more study on the use of transfer learning to the detection of various illnesses.

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